

ASSIGNMENT #6

Course: CHE 341. CHEMICAL PROCESS CONTROL
Term: Spring 2020
Professor: Dr. Michael Caracotsios
Office: EIB 244
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Your name: Christian Sanchez
Due Date: Saturday April 11th by 12:00 noon

Number of points.....100
Your score.....

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Scan your Assignment and create a PDF file
- PDF File Name: Assignment6_XY.PDF
 - X: First letter of your first name
 - Y: Your last name
 - Example: Assignment6_MCaracotsios.PDF
- Email your PDF file to: Daniel Christiansen dchris28@uic.edu and Yuechen Gao ygao54@uic.edu by the due date (Saturday April 11) and time (12:00 noon)

Note: All problems are worth 100 points. Your final score will be the average value

Problem 6A: Design of Feedback Controller for Single Loop Disturbance Rejection

The esterification of acetic acid with ethanol takes place in a water cooled jacketed Continuous Stirred Tank Reactor (CSTR) under atmospheric conditions. The reactor volume is $\{V_R = 10\text{ m}^3\}$ and the reaction mechanism and kinetic constants are given in the table below:

Reaction	$CH_3COOH + C_2H_5OH \xrightleftharpoons[k_r]{k_f} CH_3COOC_2H_5 + H_2O$
Reaction Rate	$\mathcal{R}_1 = k_f C_{AA} C_{ETOH} - k_r C_{EAC} C_{WATER}$
Forward Rate Constant	$k_f = k_{f0} \exp\left(-\frac{E_f}{R_g T_R}\right)$
Reverse Rate Constant	$k_r = k_{r0} \exp\left(-\frac{E_r}{R_g T_R}\right)$

The conditions of the feed to the reactor are given in the table below:

Feed	Data
Temperature	25°C
Pressure	1 atm
Volumetric Flow Rate	120 gpm
Acetic Acid Mole Fraction	0.495
Ethanol Mole Fraction	0.479
Ethyl Acetate Mole Fraction	0.000
Water Mole Fraction	0.026

The cooling water is available at $\{25^\circ\text{C}\}$ and a flow rate of $\{97\text{ gpm}\}$. The residence time of the water in the cooling jacket is 25% of the residence time in the reactor. The delays in the reactor temperature measurement can be modeled by a first order transfer function with gain equal to one and lag time equal to 5 min.

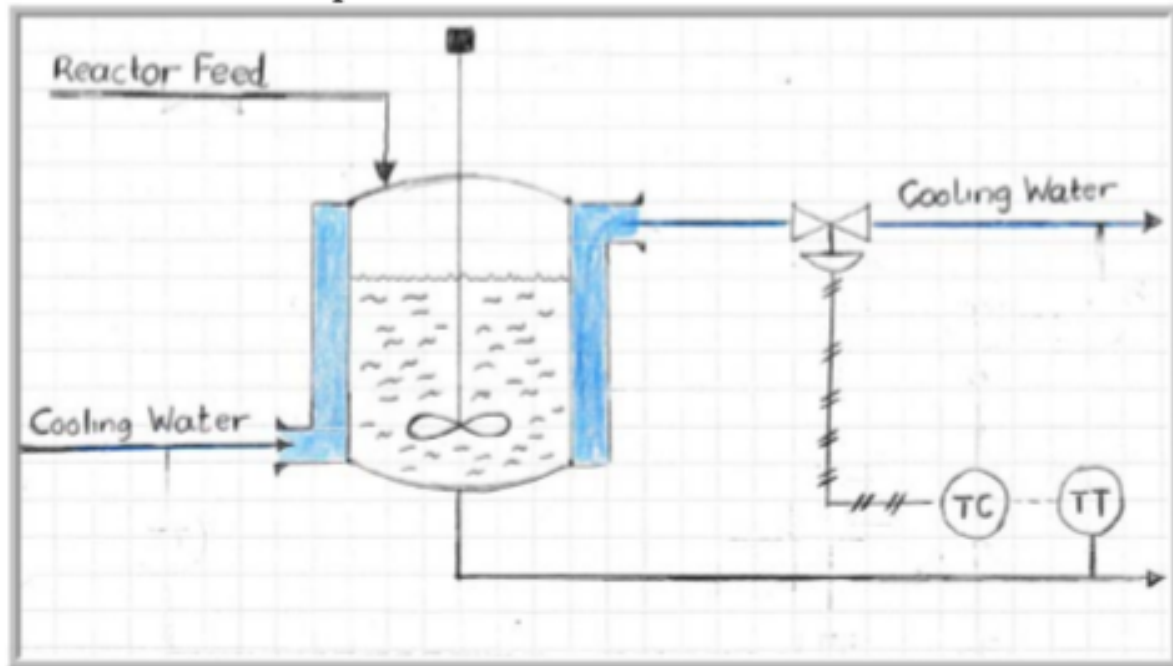
The objective is to design a feedback control loop to maintain the reactor temperature at set point by rejecting disturbances caused by changes in the cooling water jacket inlet temperature and by changes in the reactor feed temperature. Maximum temperature changes are ± 5 degrees Celsius.

A process flowsheet diagram with the feedback controller is shown below.

1. Derive by performing simulation studies(step tests) all the required FOPTD transfer functions.
2. Use the **CHE341 Calculator** to design a feedback controller that best achieves the control objective stated above

In designing the feedback controller, consider both PI and PID controllers and use the following methods to calculate the required controller parameters: Ziegler-Nichols, Cohen-Coon, Lambda tuning and Ultimate Cycling tuning. **Select the best amongst all your cases.**

Feedback Control Loop



Problem Data

Symbol	Definitions and Problem Data
$\rho = 17.8 \text{ kmol} / \text{m}^3$	Reactor feed and effluent density
$\rho_w = 54.5 \text{ kmol} / \text{m}^3$	Cooling water density
$C_{pw} = 75 \text{ kJ} / (\text{kmol} \cdot \text{K})$	Cooling water heat capacity
$UA = 60737 \text{ kJ} / (\text{hr} \cdot \text{K})$	Overall heat transfer coefficient
$C_p = 116.76 \text{ kJ} / (\text{kmol} \cdot \text{K})$	React feed and effluent heat capacity
$\Delta H_R = -29070 \text{ kJ} / \text{kmol}$	Heat of Reaction
k_{fc}	Pre-exponential factor, $\text{m}^3 / (\text{kmol} \cdot \text{s})$
k_{ra}	Pre-exponential factor, $\text{m}^3 / (\text{kmol} \cdot \text{s})$
E_{fc}	Activation energy, kJ/mol
E_{ra}	Activation energy, kJ/mol
Symbol	Kinetic Data
k_{fc}	$1.9 \times 10^8 \text{ m}^3 / (\text{kmol} \cdot \text{s})$
k_{ra}	$5.0 \times 10^7 \text{ m}^3 / (\text{kmol} \cdot \text{s})$
E_{fc}	59.50 kJ/mol
E_{ra}	59.50 kJ/mol

First the problem data was inserted into Athena tool to find the reactor temp as a function of time. The found inlet temp in the reactor is 346.4 Kelvin, now to compare with the found data from Athena the process gain, residual time, and dead time. Now using the four method equations for tuning: Z-N, C-C, Lambda, U-C

Equation for Z-N PI controller gain

$$Kc = \frac{0.9 * Kp}{2 * \tau * \theta}$$

Equation for Z-N PID controller gain

$$Kc = \frac{1.2 * Kp}{2 * \tau * \theta}$$

$$\tau_i = 3.33t_d$$

Equation for C-C PI controller gain

$$Kc = \frac{0.9}{2 * Kp} * (\frac{\tau}{\theta} + 0.092)$$

Equation for C-C PID controller gain

$$Kc = \frac{1.35}{2 * Kp} * (\frac{\tau}{\theta} + 0.185)$$

Equation for Lambda PI controller gain

$$Kc = (\frac{\tau}{Kp * \tau})$$

Equation for Lambda PID controller gain

$$Kc = (\frac{\tau}{Kp * (\tau + \theta)})$$

Equation for U-C PI controller gain

$$Kc = 0.45 * \frac{Ku}{2}$$

Equation for U-C PID controller gain

$$K_c = 0.6 * \frac{K_u}{2}$$

$$\tau_i = 0.5p_u$$

$$\tau_d = 0.125t_d$$

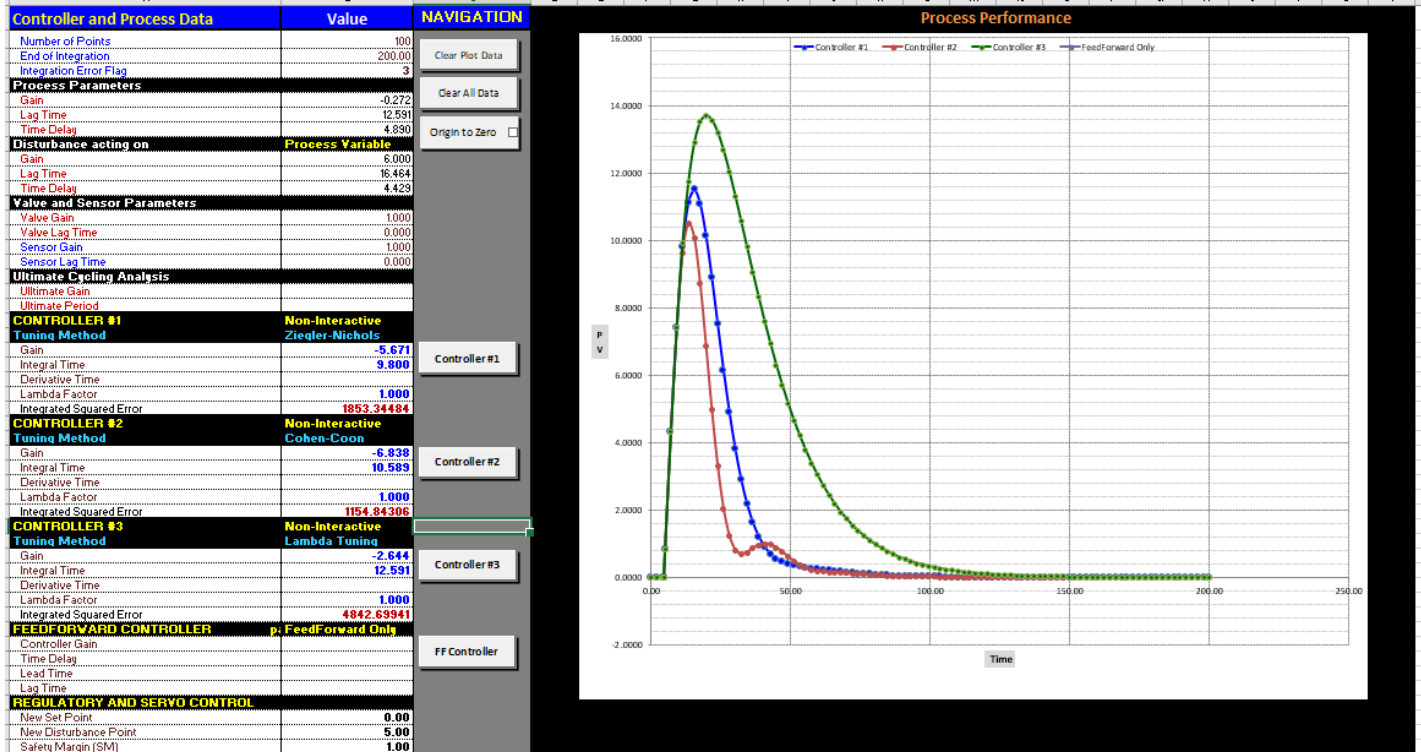


Figure 1: Process performance graph of all methods

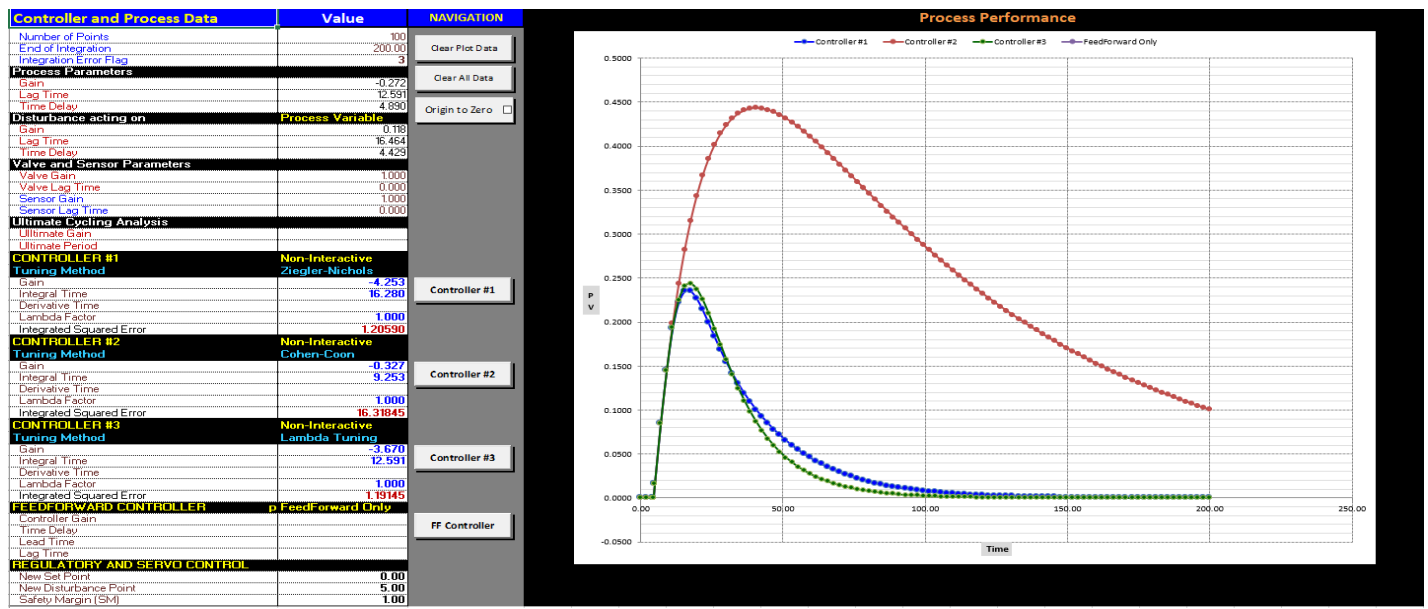


Figure 2: Process performance graph of all methods with different data

Observing the graphs above the method of choice for the figure 1 would be the C-C since it has the least amplitude of PV and with the shortest time to stabilize being 47 seconds. Furthermore, the other 2 methods are more erratic and could be harmful to the reaction. Moving on to figure 2 looking at the three methods the Lambda tuning seems to have the lowest amplitude and stabilizes in the least amount of time, thus being the method of choice.